

Tarea sección 6.4

$$3) \sec \theta = -\frac{13}{12} ; 90^\circ < \theta < 180^\circ$$

$$\sin 2\theta = 2 \sin \theta \cos \theta = 2 \left(\frac{5}{13} \right) \left(-\frac{12}{13} \right) = -\frac{120}{169}$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = \left(-\frac{12}{13} \right)^2 - \left(\frac{5}{13} \right)^2 = \frac{119}{169}$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{2(-5/12)}{1 - (-5/12)^2} = -\frac{120}{119}$$

$$\cos \theta = -12/13$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - (12/13)^2$$

$$\sin^2 \theta = 1 - 144/169$$

$$\sin \theta = \pm \sqrt{25/169}$$

$$\sin \theta = 5/13$$

$$\tan \theta = \frac{5/13}{12/13} = \frac{5}{12}$$

$$9) \sin \theta = 12/13, \quad 90^\circ < \theta < 180^\circ$$

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}} = \sqrt{\frac{1 - 5/13}{2}} = \frac{3}{\sqrt{13}}$$

$$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}} = \sqrt{\frac{1 + 5/13}{2}} = \frac{2}{\sqrt{13}}$$

$$\tan \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \sqrt{\frac{1 - 5/13}{1 + 5/13}} = \frac{3}{2}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\cos^2 \theta = 1 - (12/13)^2$$

$$\cos \theta = \pm \sqrt{25/169}$$

$$\cos \theta = \frac{5}{13}$$

$$-\frac{4}{5}$$

$$13) \cos \beta = 9/41 = \text{IV} = \text{II}$$

$$\sin^2 \beta + \cos^2 \beta = 1$$

$$\sin^2 \beta + (9/41)^2 = 1$$

$$\sin^2 \beta = 1 - (9/41)^2$$

$$\sin \beta = \pm \sqrt{1600/1681}$$

$$\sin \beta = \frac{40}{41}$$

$$\tan \frac{\beta}{2} = \pm \sqrt{\frac{1 - \cos \beta}{1 + \cos \beta}} = \sqrt{\frac{1 - 9/41}{1 + 9/41}} = \frac{4}{5}$$

$$\tan \beta = \frac{40/41}{9/41} = \frac{40}{9}$$

15) a) $\cos 67^{\circ} 30'$

$$\cos 135/2 = \sqrt{1+\sqrt{3}}/2 = \sqrt{2-\sqrt{3}}/2$$

b) $\sin 15^{\circ} = \frac{1}{2} \sqrt{2-\sqrt{3}}$

c) $\tan 15^{\circ} = \sqrt{2}-1$

21) $\sin^2 \frac{x}{2} = \frac{\sin^2 x}{2(1+\cos x)}$

$$\sin^2 \frac{x}{2} = \frac{1-\cos^2 x}{2(1+\cos x)}$$

$$\sin^2 \frac{x}{2} = \frac{(1-\cos x)(1+\cos x)}{2(1+\cos x)}$$

$$\sin^2 \frac{x}{2} = \frac{1-\cos x}{2}$$

$$\sin^2 \frac{x}{2} = \sin^2 \frac{x}{2}$$

33) $2 \sin^2 2t + \cos 4t = 1$

$$2 \sin^2 2t + \cos^2 2t - \sin^2 2t = 1$$

$$\sin^2 2t + \cos^2 2t = 1$$

$$1 = 1$$

$$41) \sin^2 2x$$

$$\frac{3}{8} - \frac{1}{2} \cos 4x = \frac{1}{8} \cos 8x$$

$$42) \sin \frac{1}{2} u + \cos u = 1$$

$$\sin \frac{1}{2} u + \sqrt{1 - \cos^2 u} = 1$$

$$\sqrt{1 - \sin^2 u} = 1 - \sin \frac{1}{2} u$$

$$1 - \sin^2 u = (1 - \sin \frac{1}{2} u)^2$$

$$2 \sin \frac{1}{2} u - \sin^2 \frac{1}{2} u - \sin^2 u = 0$$

$$0, \frac{\pi}{3}, \frac{5\pi}{3}$$